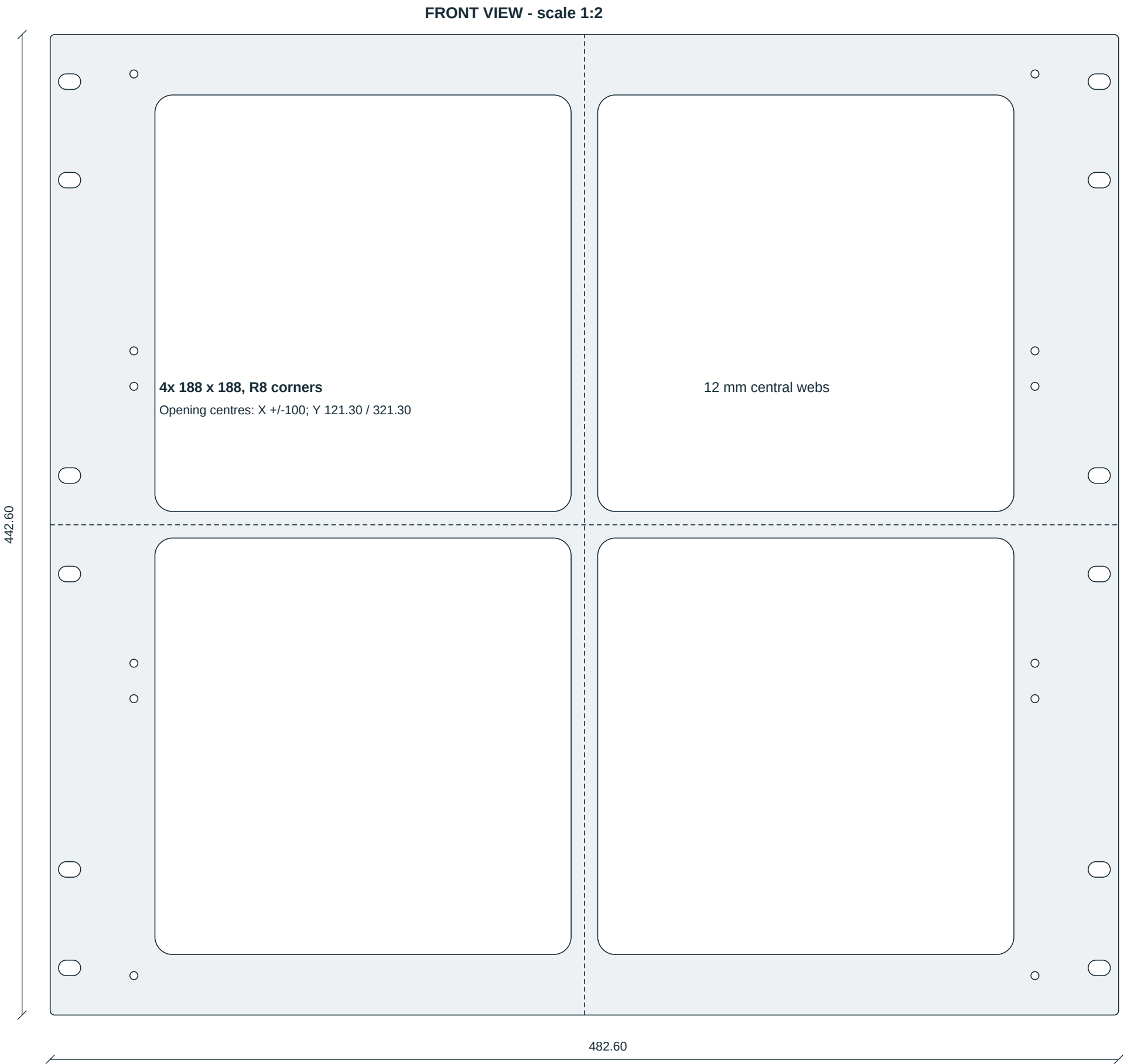


SUPERNOVA 1260 / RACK PLATE R1

304 / EN 1.4301 stainless steel | 3.00 mm sheet | Flat part, no bends | All dimensions in mm



A / RADIATOR ATTACHMENT

12x diameter 3.60 THROUGH; no countersinks.

Two columns: X = -203.50 and +203.50.

Each column uses these Y coordinates:

A1	17.800
A2	142.800
A3	158.800
A4	283.800
A5	299.800
A6	424.800

B / RACK FIXINGS

12x horizontal slots, 10.00 x 7.00, R3.50 ends.

Two columns: X = -232.55 and +232.55.

Each column uses these Y coordinates:

B1	21.275
B2	65.725
B3	199.075
B4	243.525
B5	376.875
B6	421.325

C / OUTLINE AND SECTION

4x outer corners R2.00. Thickness 3.00 +/-0.10.

No tapped holes. All cuts pass through the plate.

Sheet flatness: 0.5 mm overall target.

Profile / hole centre dimensions: +/-0.15 mm.

Hole / slot widths: +0.15 / 0 mm.

Deburr both faces and all cut edges, 0.2-0.3 mm.

Satin finish; no markings on the finished part.

RADIATOR DATUM

422 x 441 body centred at X=0, Y=221.30.

Stock mesh centre: X=0, Z=212.00.

Transform: plate Y = mesh Z + 9.30.

Ports face top or bottom. Fittings not in envelope.

Matching STEP and cut-only DXF: output/radiator-R1/

SUPERNOVA 1260 / RACK PLATE R1

Load estimate, assembly intent and interface verification | R1 prototype design

WEIGHT AND STATIC LOAD

Published radiator net weight	4.225 kg
Operator allowance	2.000 kg
Load carried by radiator attachment screws	6.225 kg / 61.0 N
New plate, CAD volume x 7900 kg/m3	1.697 kg
Total estimated rack load	7.922 kg / 77.7 N

No credit is taken for removing the original front fan plate.

The 2 kg is a combined allowance, not a measured coolant/fan BOM.

Additional hardware or a pump/reservoir beyond that allowance adds load.

STATIC REACTION SCREEN - NOT A LOAD RATING

Payload centre of gravity assumed 75 mm behind the rack plate.

Payload overturning moment: 4.58 Nm.

Four outermost rack screws: average vertical shear 19.4 N each.

Top rack-screw pair total tension from moment: 11.4 N.

Twelve radiator screws: average vertical shear 5.1 N each.

These averages assume symmetric sharing; clearance holes may share unevenly.

Doubling 1.5 to 3 mm gives 8x bending stiffness for an identical plate profile.

This does not establish assembly deflection, pull-out strength or shock resistance.

GEOMETRY CHECKS

One valid solid; STEP re-import volume and extents checked.

DXF: 1 outline + 4 openings + 12 slots + 12 holes; closed cut contours.

Plate airflow opening area: 141156 mm2; 88.2% of a 400 mm square.

Minimum straight hole-to-air-opening ligament: 7.7 mm.

Rack slot to outer plate edge: 3.75 mm; use flat washers.

12 attachment centres cross-checked against the manufacturer STL pilots.

ASSEMBLY

Replace the front fan plate with R1; retain the opposite fan plate.

Four 200 mm rear fans shown in the review model; nine 140 mm fans remain an option with the matching opposite plate.

Use all 12 radiator retention points. A4 M3 pan-head screws and suitable flat washers; maximum washer OD 7 mm.

Starting candidate: M3 x 6, giving 3 mm reach before a washer.

Confirm actual required engagement and safe depth before use.

Do not use the supplied M3 x 30 fan screws on the bare plate.

Rack: M6 screws / cage nuts where compatible with actual rails.

Proposed installation: 4 per side, rows B1/B3/B4/B6.

B2/B5 are additional optional rack fixing positions.

The static screen conservatively uses only four corner screws.

CHECK BEFORE FABRICATION / INSTALLATION

Confirm manufacturer revision, mounting pattern and seating faces.

The catalogue mesh is corroboration, not a toleranced interface drawing.

Apertures remove fan-only holes; actual structural side mounts remain.

Check 3 mm plate flatness and contact against the radiator side rails.

Verify the thin radiator rails/inserts can carry the intended rack load.

Review top/bottom elbows against host/base and plate edge.

No wet joint, radiator pressure boundary or manifold part is modified.

SOURCE REFERENCES

Alphacool 14351 datasheet v1.007, 03.2026, pp2 and 4-6.

Alphacool 3D Centre: 14351_0.stl (checked 15 September 2026).

Outokumpu Core range: 304 density 7.9 kg/dm3, E = 200 GPa.

Full URLs, mesh hash and calculations accompany the CAD package.